



ARP

CONFIGURATION

USING CISCO PACKET TRACER



The invisible handshake of every LAN.



01

ARP REQUEST

PC-0 broadcasts to **FF:FF:FF:FF:FF:FF**. All LAN devices receive the frame asking "Who has 192.168.1.3?"

BROADCAST



02

ARP REPLY

PC-3 recognises its IP and replies directly to PC-0 with its MAC **0040.0bea.2137** — unicast only.

UNICAST

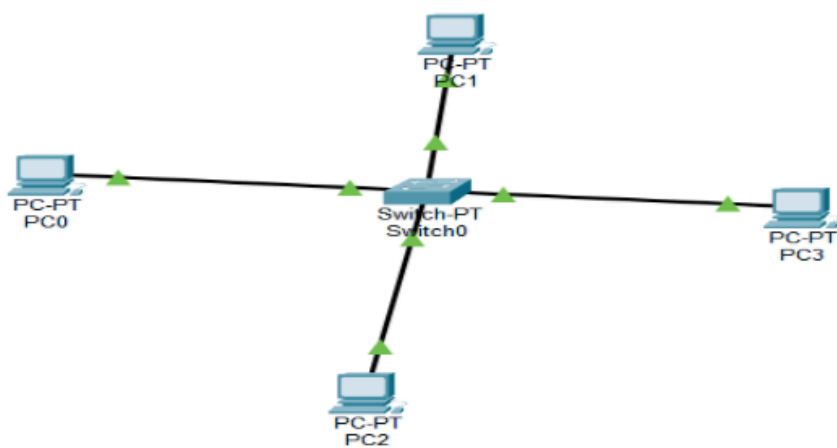


03

ARP CACHE

IP→MAC mapping stored as a **dynamic entry**. All future packets use cached MAC — zero broadcasts needed.

DYNAMIC · AUTO-EXPIRE



DEVICE CONFIGURATION

DEVICE	IP ADDRESS	PORT
PC-0	192.168.1.1	Fa0/1
PC-1	192.168.1.2	Fa0/2
PC-3	192.168.1.3	Fa0/3
PC-2	192.168.1.4	Fa0/4

```
C:\> arp -a
192.168.1.3 -> 0040.0bea.2137 dynamic
```

ADVANTAGES

- Fast LAN Communication — automatic MAC resolution, near-instant packet delivery.
- Zero Manual Config — plug-and-play; no static MAC tables required.
- Transparent — runs silently inside the OS networking stack.

LIMITATIONS

- LAN-Only Scope — does not cross routers or VLAN boundaries.
- ARP Spoofing — fake replies can poison cache; enables MitM attacks.
- No Authentication — any host can claim any IP's MAC address.

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